

Project Report

Cosmetic Insights: A Tableau-Based Analysis of Skincare Trends and Consumer Preferences

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1. Introduction

1.1 Project Overview

Cosmetic Insights is a Tableau-based data analytics project that examines a real-world cosmetics product dataset of 1,472 products from 116 brands across 6 categories. The project builds 18 interactive worksheets and a guided storyboard to help users compare products by brand, price, customer rating, category, and skin-type suitability.

1.2 Purpose

The purpose of this project is to give skincare buyers and cosmetic brand/retail analysts a centralized, visual way to compare products on the dimensions that matter most, replacing scattered, unstructured listings with an interactive dashboard and data-backed insights.

2. Ideation Phase

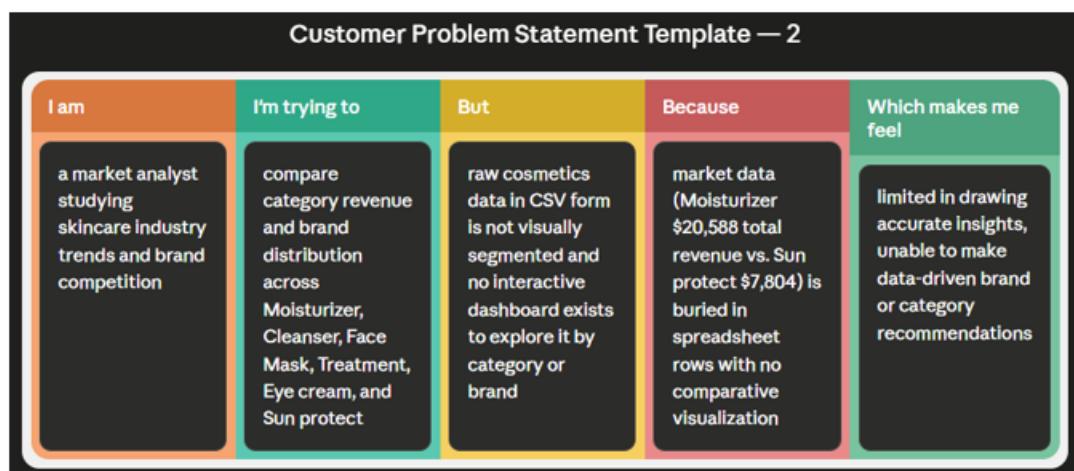
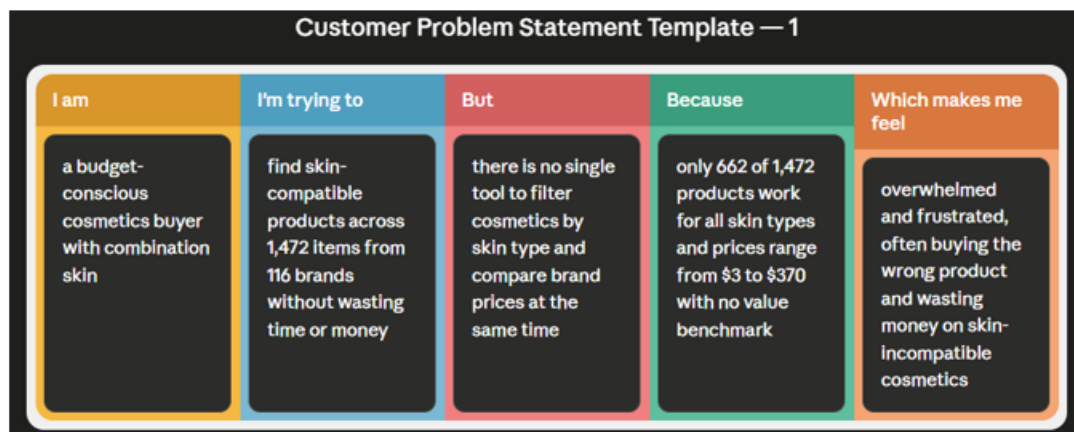
2.1 Problem Statement

Buyers and brand analysts lack a centralized, visual way to compare cosmetic products on the dimensions that matter most — skin-type suitability, price, brand, category, and customer rating. Existing product listings present this information in an unstructured, hard-to-compare format, leading to uncertainty and reliance on guesswork or marketing claims (full Customer Problem Statement table in Section 2 of the project repository).

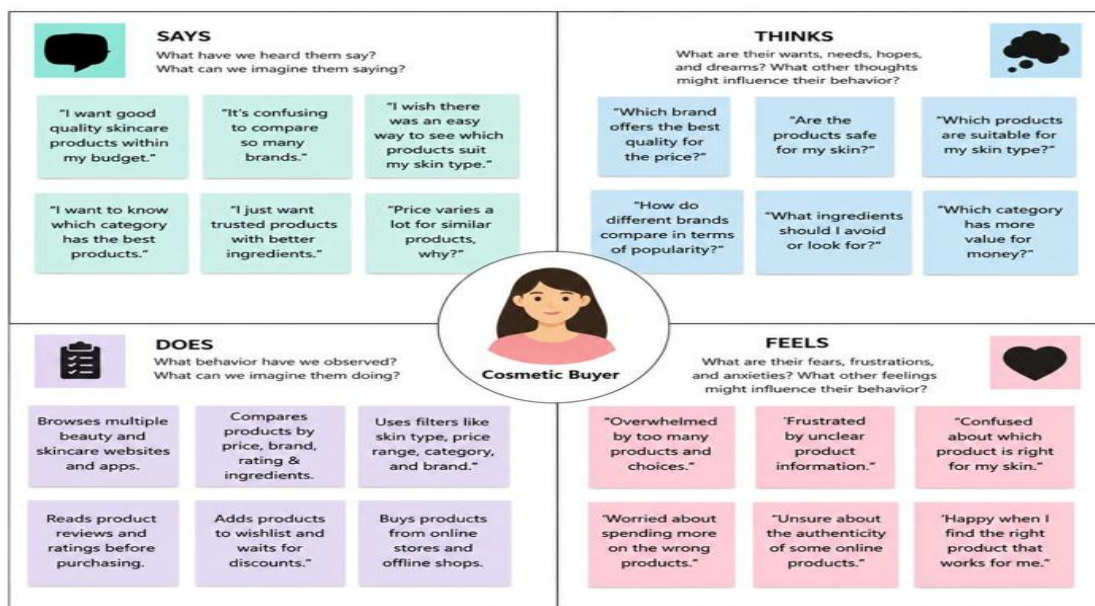
Problem Statement Table

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A skincare buyer choosing products for my skin type and budget	find a product that suits my skin type (Dry / Oily / Sensitive / Combination / Normal) without overspending	product listings are scattered across sites and don't let me compare brands, price, and rating together	there is no single, centralized, visual tool covering 1,472 products across 116 brands and 6 categories	overwhelmed and unsure, and I end up relying on guesswork or brand marketing instead of data
PS-2	A cosmetic brand / retail analyst	understand how my brand's product range and pricing compare against 115 other competing brands	the raw catalogue data has no visual, comparative view of brand performance, pricing, or category share	category and brand-level patterns are buried in 1,472 unstructured product rows	unable to quickly benchmark performance and respond to market gaps

Example: Cosmetic Insights: A Tableau-Based Analysis of Skincare Trends and Consumer Preferences



2.2 Empathy Map Canvas



An empathy map was built around the persona of a skincare buyer / brand analyst, covering what they Think & Feel, Hear, See, and Say & Do, along with their Pain and Gain points. The key insight: the core unmet need is a way to see existing product data clearly and compare it on skin type, price, rating, and brand — not more raw data (full canvas in Section 2 of the project repository).

2.3 Brainstorming

The team followed a structured How-Might-We -> Generate -> Group -> Prioritize process. Ideas were grouped into four themes (Brand & Category Analysis, Price Analysis, Skin-Type Suitability, Price vs. Rating) and placed on an importance-vs-feasibility grid, carrying forward high-priority ideas such as top-10 brand views, skin-type breakdowns, and the price-vs-rank scatter plot into development.

3. Requirement Analysis

3.1 Customer Journey Map

The journey was mapped across five stages — Discover, Explore, Compare, Decide, Reflect — tracing how a buyer moves from realizing they need a product to confidently selecting one, and how an analyst benchmarks brand performance. Each stage was mapped to specific dashboard worksheets (full swimlane map in Section 3 of the project repository).

3.2 Solution Requirement

Functional requirements (FR-1 to FR-7) cover data import, category analysis, brand analysis, skin-type suitability analysis, price-vs-rating analysis, all-skin-type brand recommendations, and storyboard presentation. Non-functional requirements (NFR-1 to NFR-6) cover usability, security, reliability, performance, availability, and scalability (full tables in Section 3 of the project repository).

3.3 Data Flow Diagram

Data flows from the raw cosmetics.csv file (1,472 rows) through Excel-based cleaning, into a single Tableau data source, which feeds all 18 worksheets, which are assembled into Story 1 and published as a packaged workbook / Tableau Public link (full DFD and User Story table in Section 3 of the project repository).

3.4 Technology Stack

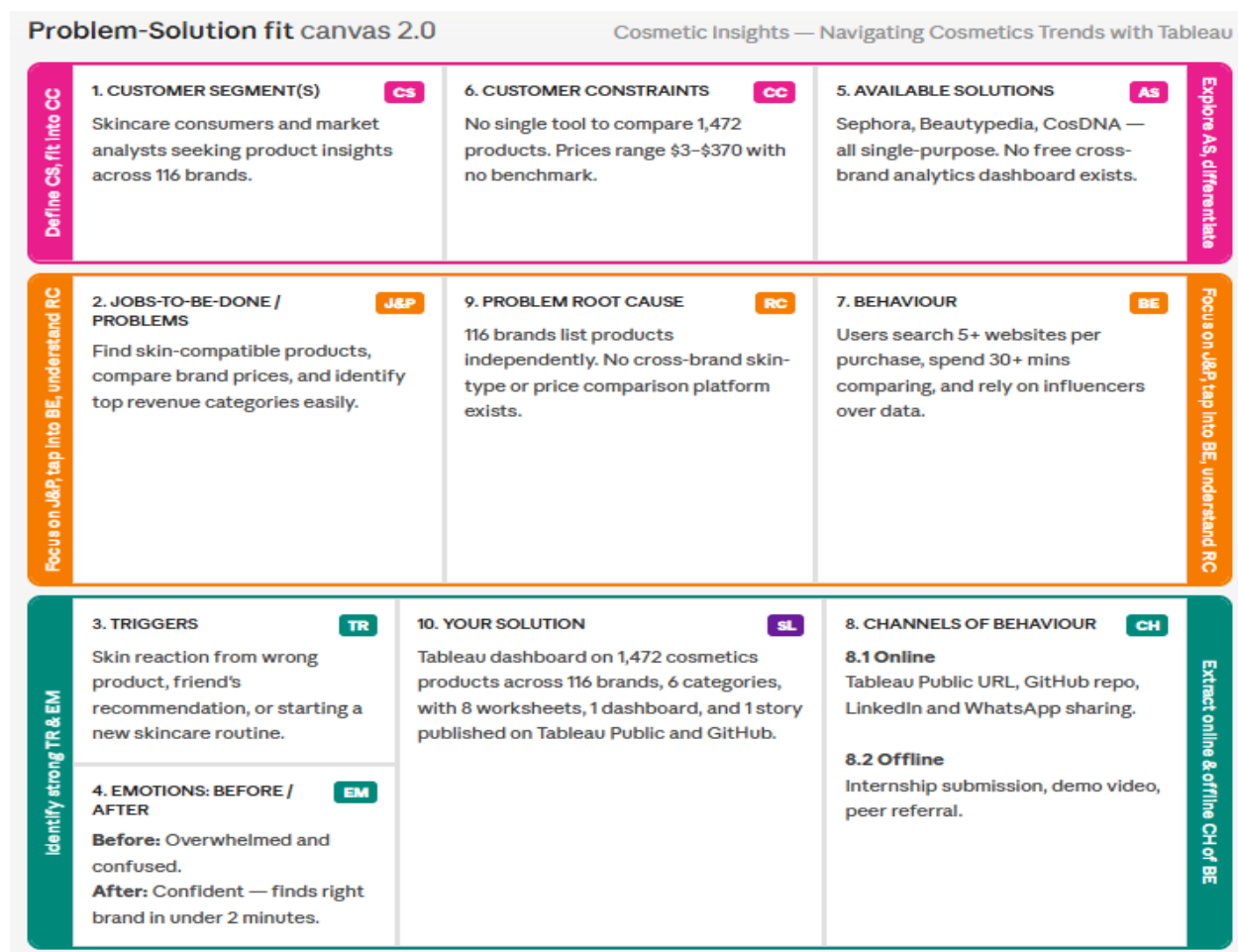
The stack consists of CSV/Excel for data storage and cleaning, Tableau Desktop / Tableau Public for visualization, the .twbx packaged workbook format for distribution, and GitHub for version control and submission (full component and characteristics tables in Section 3 of the project repository).

4. Project Design

4.1 Problem Solution Fit

The 10-box Problem-Solution Fit canvas confirms the solution fits the validated problem: the Tableau workbook addresses the customer segment's job-to-be-done (compare 1,472 products by skin type, price, and rating),

removes the root cause (no aggregated visual view existed), and matches observed buyer behavior (quick visual comparison instead of manual research).



4.2 Proposed Solution

The proposed solution is an interactive Tableau workbook (18 worksheets + storyboard) built on the cosmetics dataset. Its novelty lies in unifying brand, price, rating, and skin-type comparison in one consistent, filterable view, with direct social impact for buyers with Sensitive or Dry skin who have comparatively fewer suitable product options in the catalogue.

4.3 Solution Architecture

The architecture flows from the raw data layer (cosmetics.csv) through a preparation layer (Excel cleaning), into a single Tableau data source, an analysis layer of 18 worksheets, a presentation layer (Story 1), and a distribution layer (Tableau Public / .twbx / screenshots) (full architecture diagram in Section 4 of the project repository).

5. Project Planning & Scheduling

5.1 Project Planning

Work was organized into 2 sprints totaling 20 story points across 10 user stories, covering data preparation, category analysis, brand analysis, skin-type analysis, price-vs-rating analysis, storyboard assembly, and

documentation. Both sprints completed on schedule with an average velocity of 4 story points per day (full backlog and sprint tracker in Section 5 of the project repository).

6. Functional and Performance Testing

6.1 Performance Testing

All 1,472 rows rendered correctly into the Tableau data source. Data preprocessing (category and skin-type flag standardization) was verified, filters and calculation fields (distinct count of Name, SUM(Price), COUNTD(Brand), AVG(Rank)) were validated, and both the 8-panel dashboard and the 5-point Story 1 storyboard rendered and behaved as expected (full test parameter table in Section 7 of the project repository).

7. Results

7.1 Output Screenshots

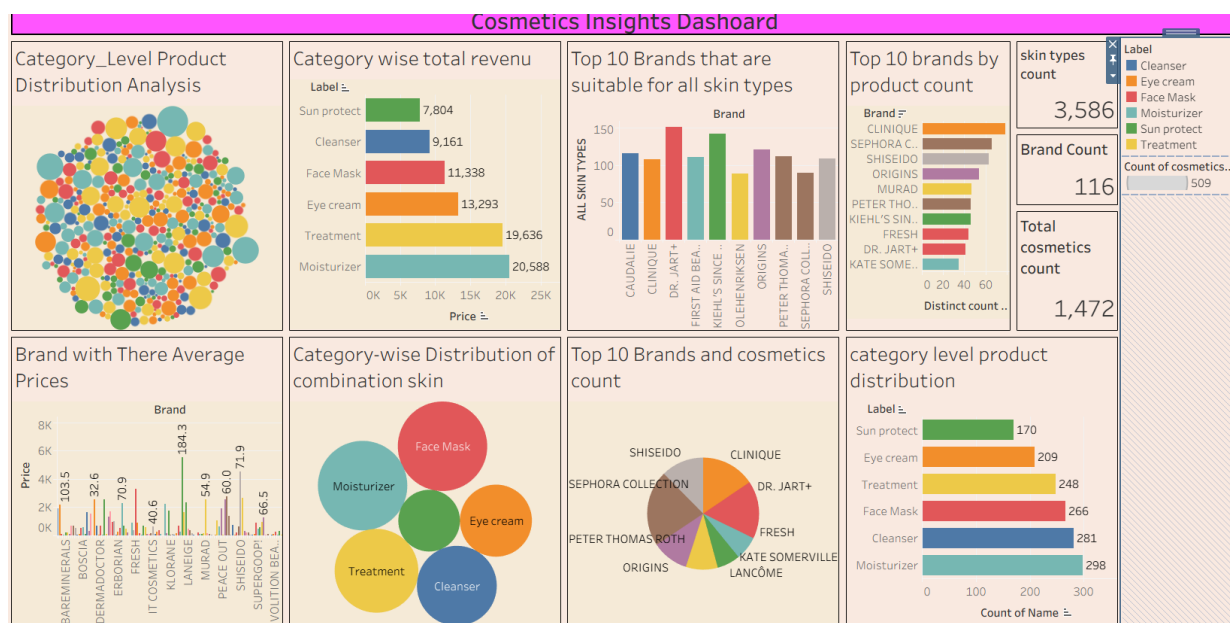


Figure 1: Cosmetics Insights Dashboard

Category Distribution: Moisturizers form the largest category with 298 products, followed by Cleansers (281), Face Masks (266), Treatments (248), Eye creams (209), and Sun protection products (170).

Story 1

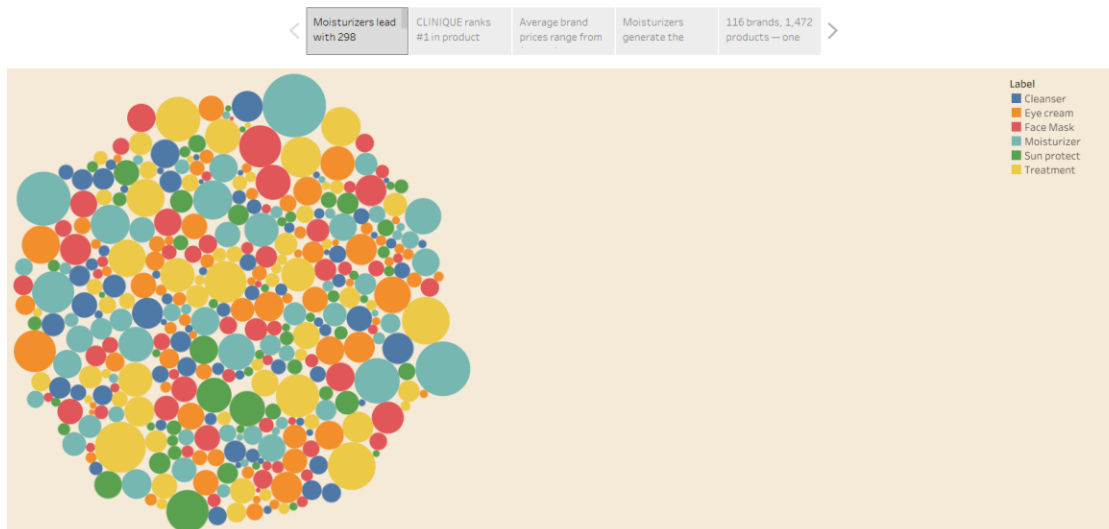


Figure 2: Category distribution (package-bubble visualization)

Brand Analysis: Across 116 brands, CLINIQUE leads with 79 products, followed by SEPHORA COLLECTION (66), SHISEIDO (63), ORIGINS (54), and MURAD (47).

Story 1

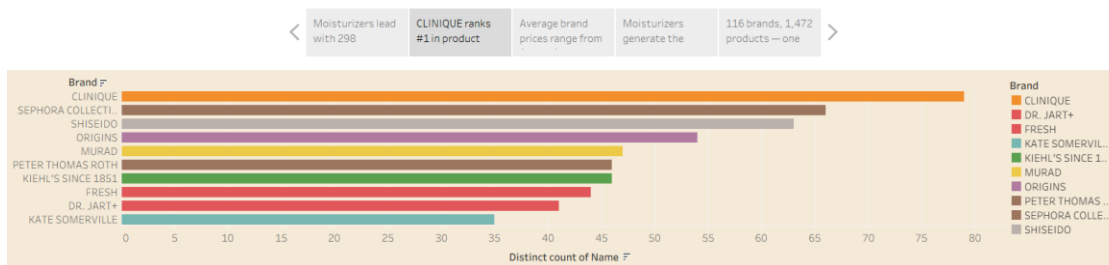


Figure 3: Top 10 brands by distinct product count

Price Analysis: Prices range from \$3 to \$370, with an average of \$55.58 across the catalogue and considerable variance across brands.

Story 1

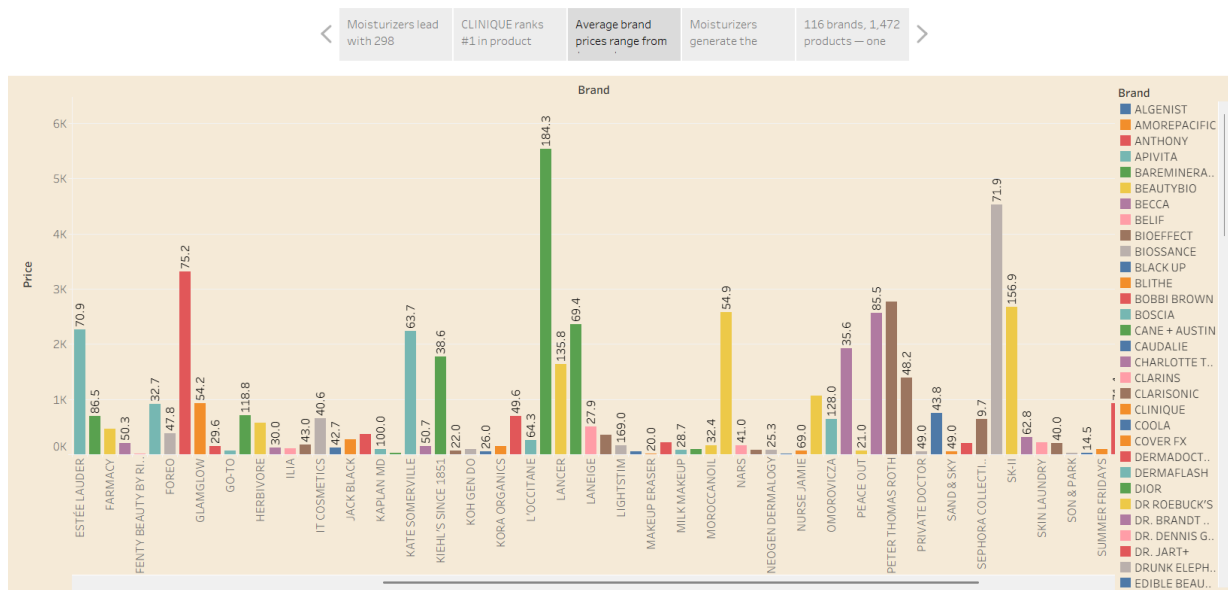


Figure 4: Average brand prices across the full brand list

Category Revenue: Moisturizer (\$20,588) and Treatment (\$19,636) categories generate the highest cumulative price-based revenue, while Sun protect (\$7,804) is the lowest.

Story 1

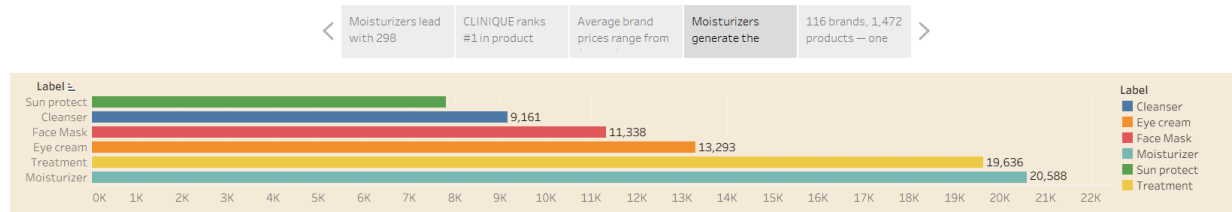


Figure 5: Category-wise total revenue (price-based)

Pie chart showing product share distribution among the top 10 cosmetics brands, with CLINIQUE holding the largest slice followed by SEPHORA COLLECTION and SHISEIDO.

Moisturizers lead with 298 CLINIQUE ranks #1 in product Top 10 Brands by Cosmetics Count Average brand prices range from Combination Skin Category Moisturizers generate the Price vs Rank Analysis 116 brands, 1,472 products — one

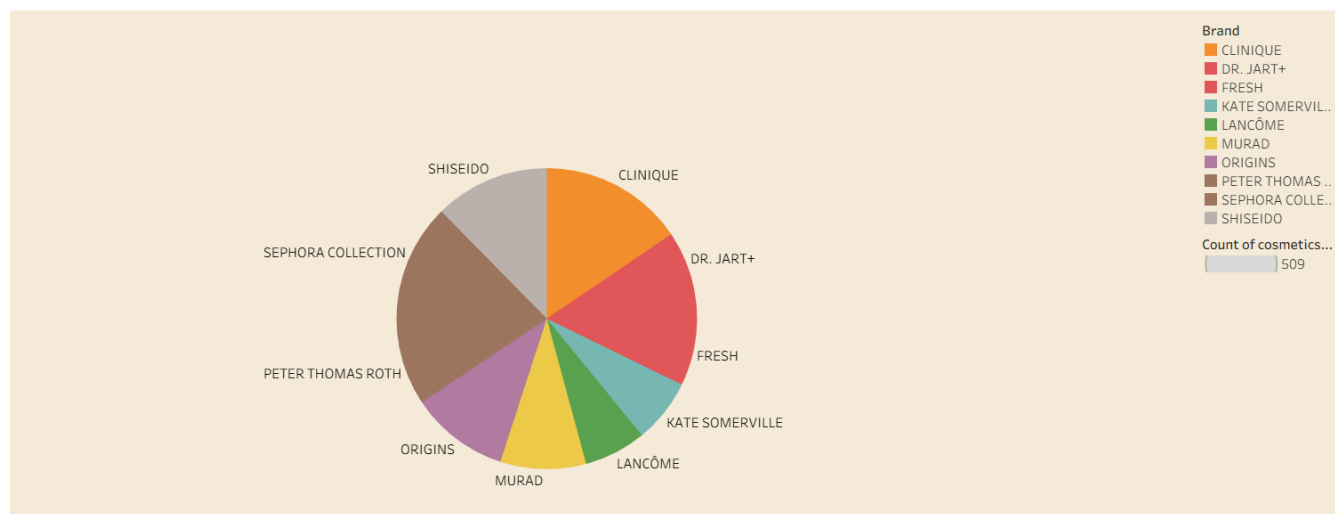


Figure 6: Top 10 Brands Share by

Packed bubble chart displaying category-wise product distribution for combination skin, where Face Mask leads as the largest bubble followed by Moisturizer and Cleanser.

Story 1

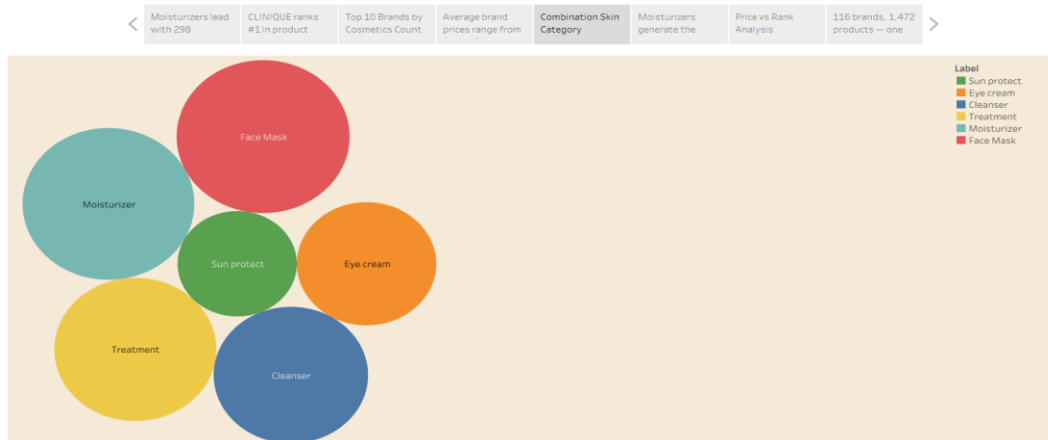


Figure 7: Category Distribution for Combination Skin Type

Scatter plot comparing product price against consumer rank across all 116 brands, revealing no strong correlation — high-ranked products exist across both low and high price ranges.

Story 1

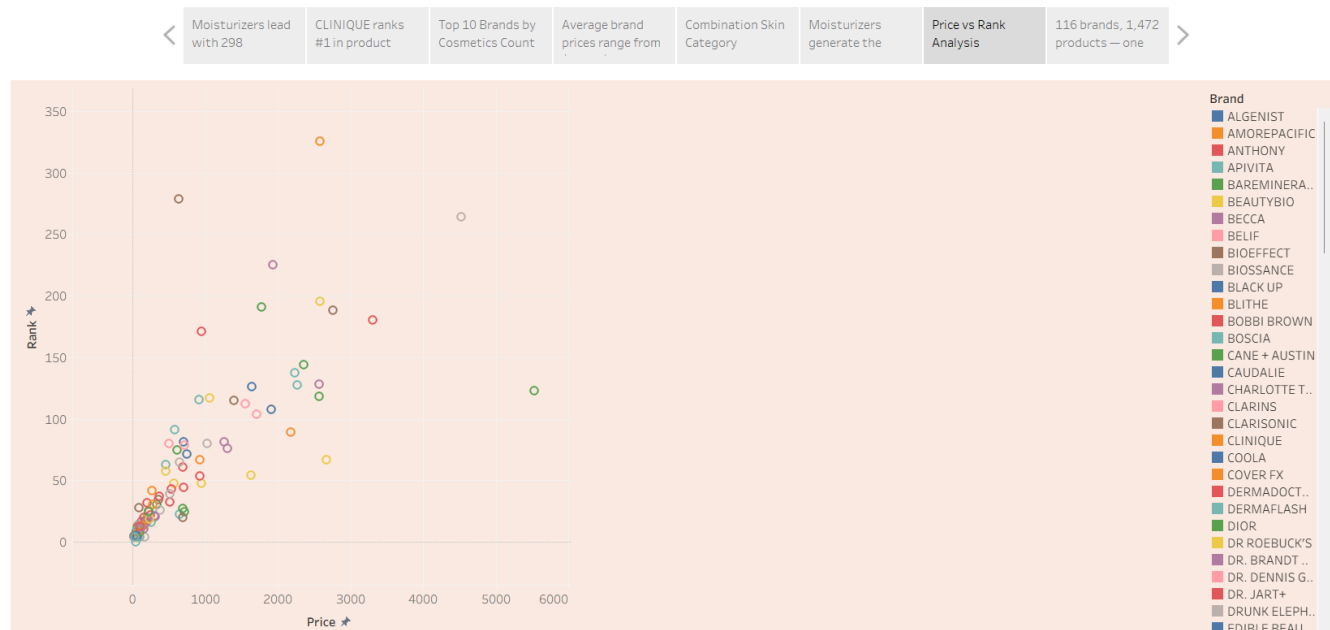


Figure 8: Price vs Rank Scatter Across All Brands

8. Advantages & Disadvantages

8.1 Advantages

- Consolidates 1,472 scattered product records into one interactive, filterable workbook.
- Lets buyers quickly identify products suited to their skin type and budget.
- Gives brand analysts a fast way to benchmark catalogue size, pricing, and category share against 115 competitors.
- Single data source keeps all 18 worksheets consistent and easy to maintain.

8.2 Disadvantages

- Static snapshot dataset — does not reflect real-time price or inventory changes.
- No ingredient-level safety or allergen analysis despite the Ingredients field being available.
- No personalization layer — recommendations are general, not tailored to an individual user profile.

9. Conclusion

This project successfully transforms a raw, 1,472-row cosmetics dataset into an interactive, decision-ready Tableau workbook. The analysis confirms that the product catalogue is broad across categories, moderately concentrated among a few large brands, broadly priced, and unevenly supportive of sensitive skin (756 products) compared to other skin types (904-966 products). These insights directly answer the questions raised

during the Ideation and Requirement Analysis phases and demonstrate how visual analytics can simplify decision-making for both consumers and brand analysts.

10. Future Scope

- Incorporate ingredient-level analysis (e.g., flagging common allergens or actives) using the existing Ingredients field.
- Add a recommendation layer that suggests products based on a user's stated skin type and budget.
- Expand the dataset with time-series pricing or review data to track trends over time.
- Publish the dashboard on Tableau Public with embedded filters for broader public access.

11. Appendix

Dataset Link

Dataset: cosmetics.csv (1,472 records, 116 brands, 6 categories) — included in Section 6, Project Executable Files, of the project repository.

GitHub & Project Demo Link

GitHub Repository:

<https://github.com/Karun-kumar-123/Cosmetic-Insights-Navigating-Cosmetics-Trends-and-Consumer-Insights-with-Tableau>

Project Demo Video:

<https://drive.google.com/file/d/1ACu1djEvpOwwD0MLxXM8iesJXVFPNY4K/view?usp=sharing>